

S&P 500 Factor Investing

- Group participants: add names before submission.
- Assignment setting adapted from BIST100 to S&P 500.
- Data window: all available observations through May 2026.
- Required factors: ROE, P/E, momentum, and trend.

Paper Summary

- Han, Zhou, and Zhu propose a trend factor using moving averages across multiple horizons.
- Normalized moving averages combine short-term reversal, intermediate momentum, and long-term reversal information.
- The paper forms top-minus-bottom portfolios from expected returns predicted by trend signals.
- Our project keeps the idea but adapts the index input to $\wedge\text{GSPC}$.

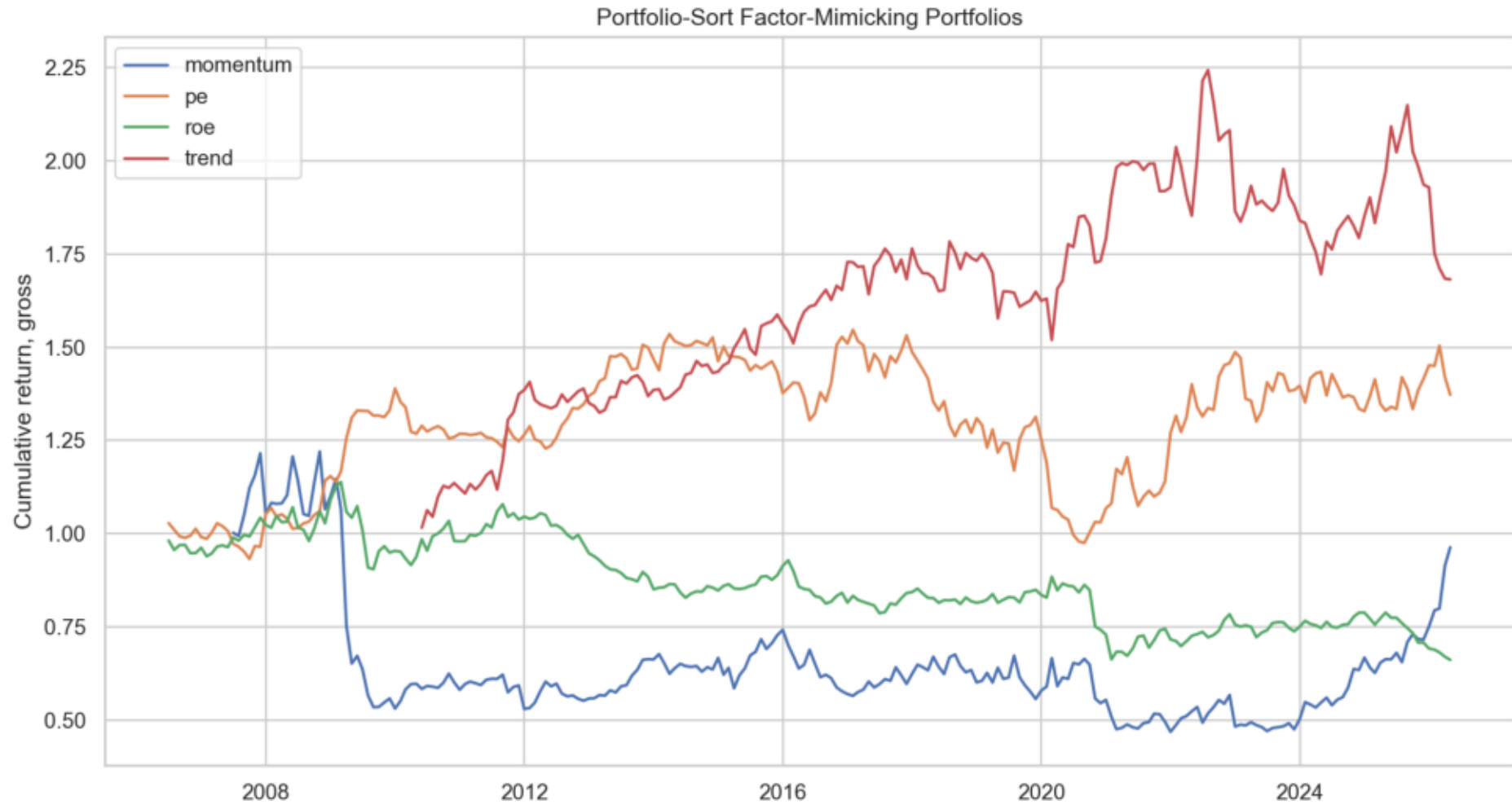
Data And Signals

- Stocks: current S&P 500 constituent panel from Tiingo data.
- Benchmark and trend-regression index: Yahoo Finance ^GSPC.
- Fundamentals are joined by public availability date to avoid look-ahead bias.
- All factors are month-end, winsorized, z-scored, and lagged before trading.

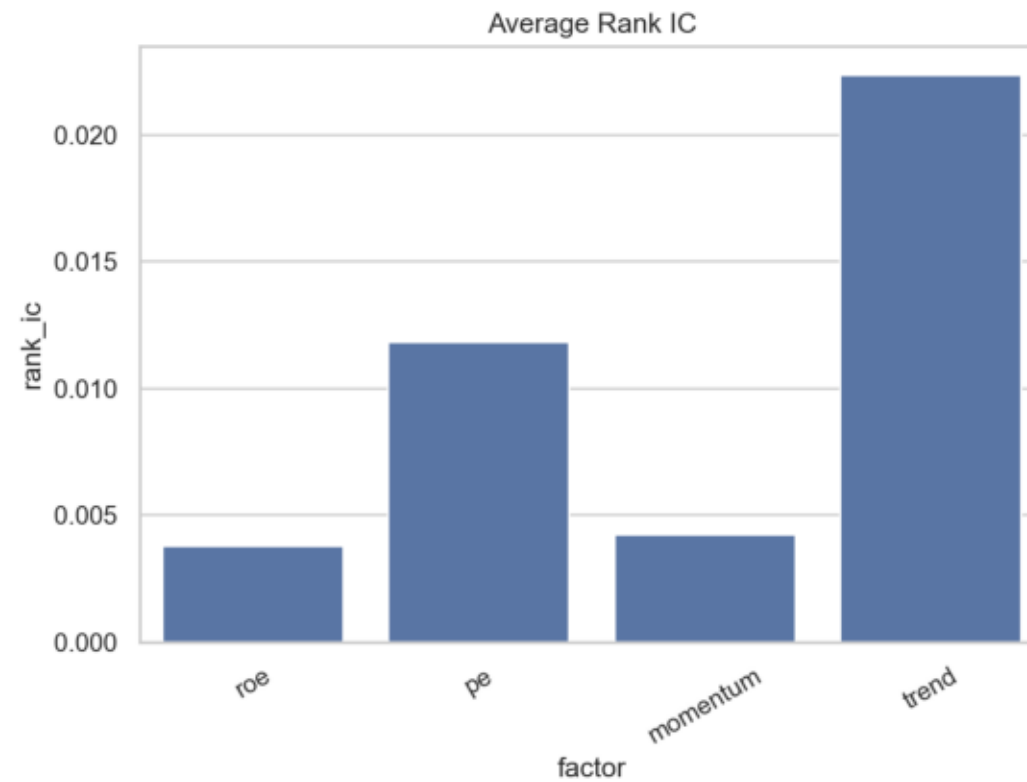
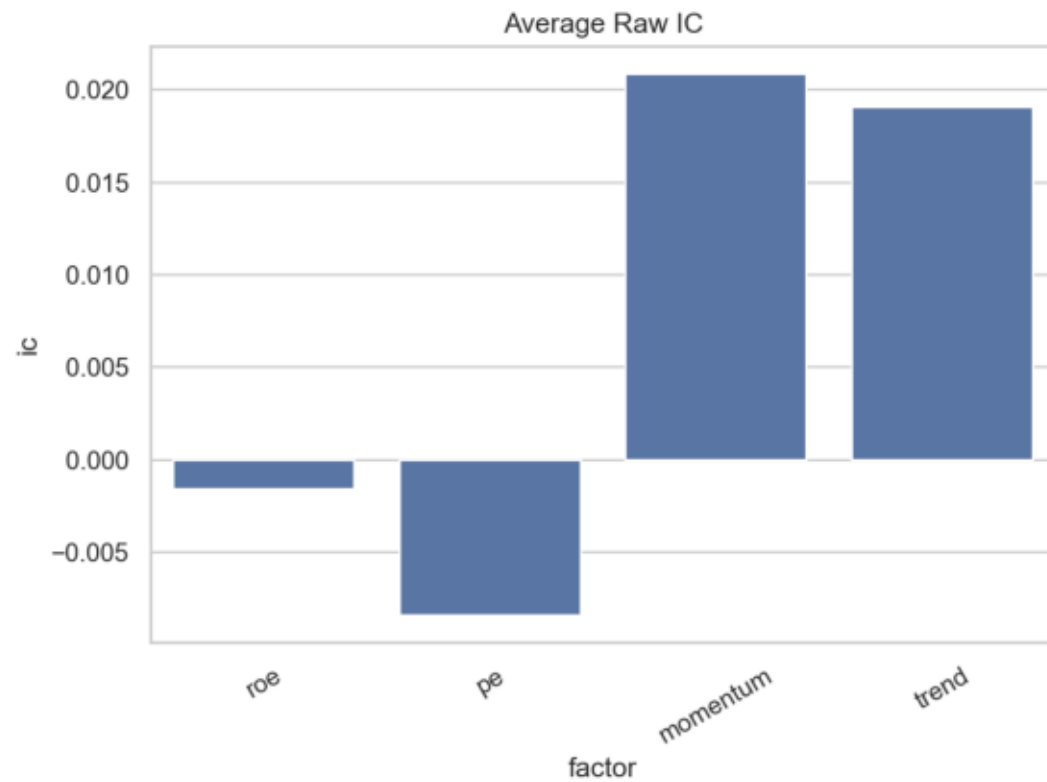
Backtest Design

- Initial cash: 1,000,000.
- FixedCashSizer: 100,000 per trade.
- Commission: 0.
- Base strategy: long-only top 10 composite factor stocks, monthly rebalance, market orders.
- Improved 1: same base strategy with expanding no-lookahead trend regression.
- Improved 2: improved 1 plus 10% stop-loss and 20% take-profit.
- Improved 3: improved 2 plus rolling rank-IC factor weights with shrinkage and caps.

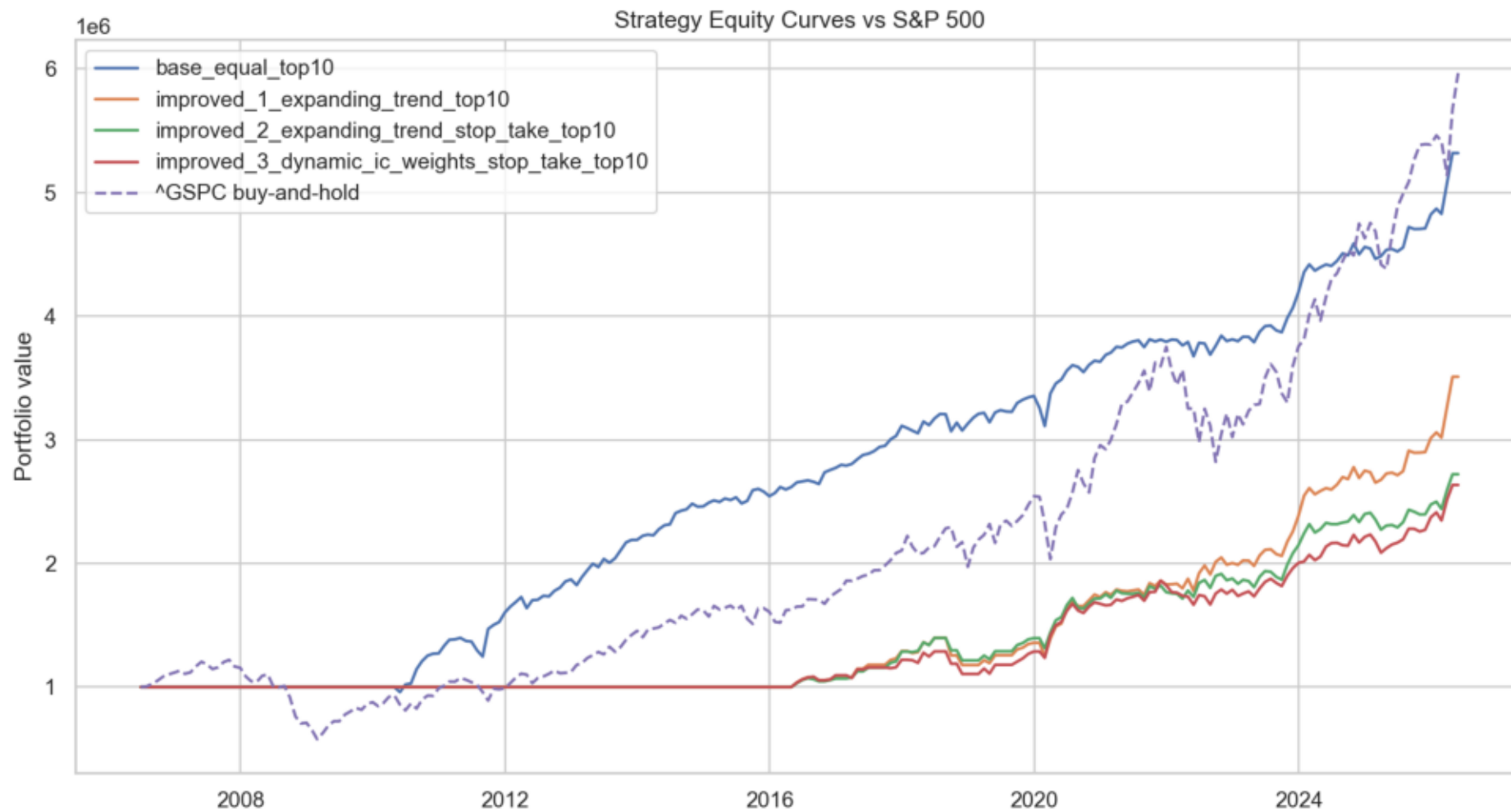
Factor Portfolio Performance



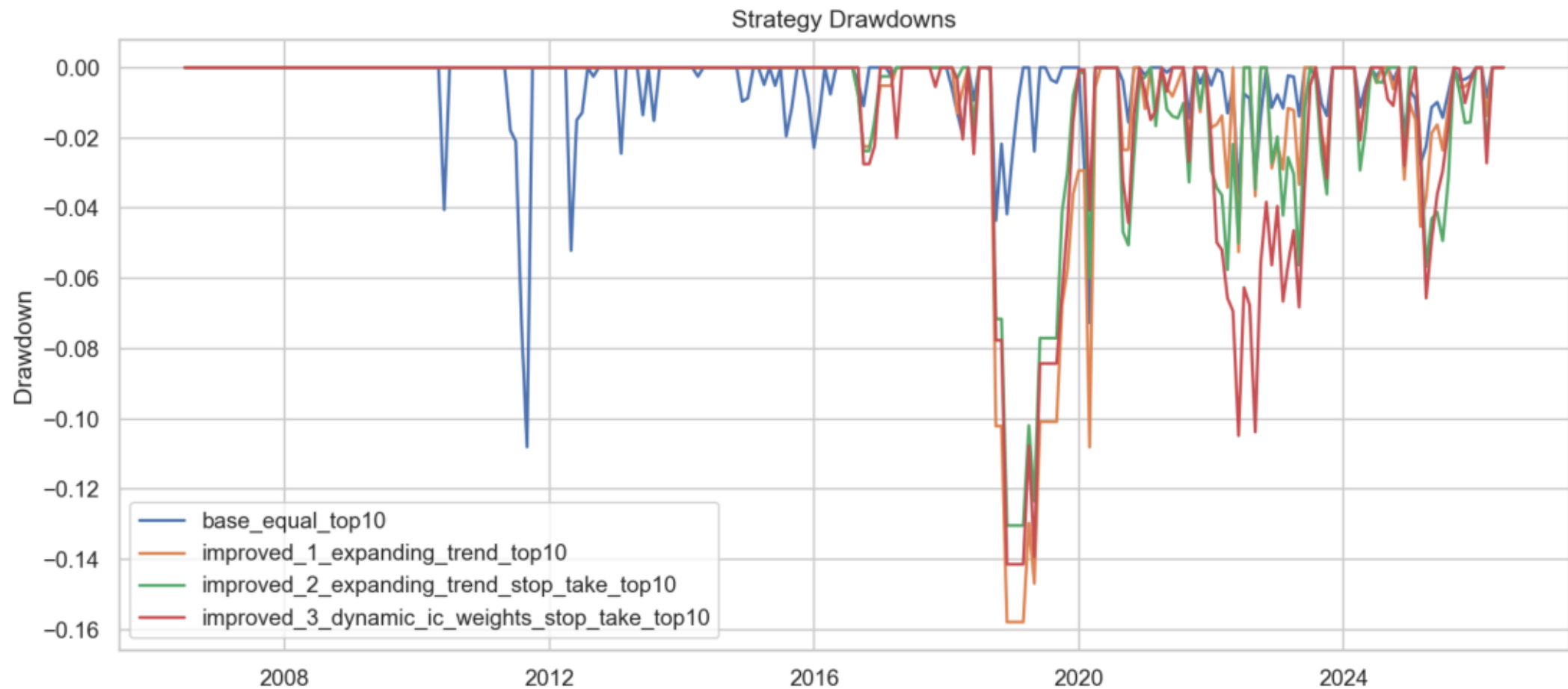
Information Coefficients



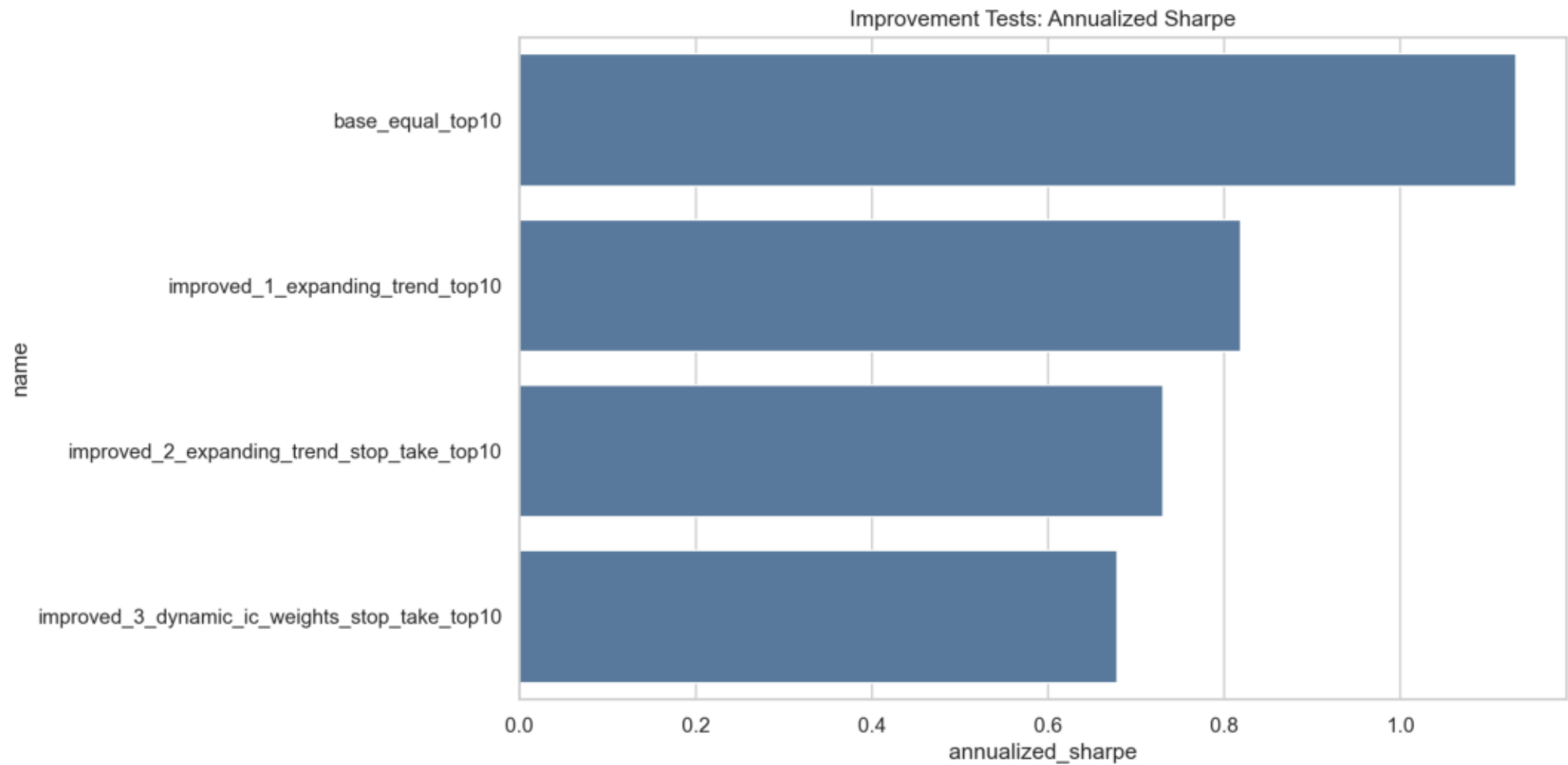
Strategy vs Benchmark



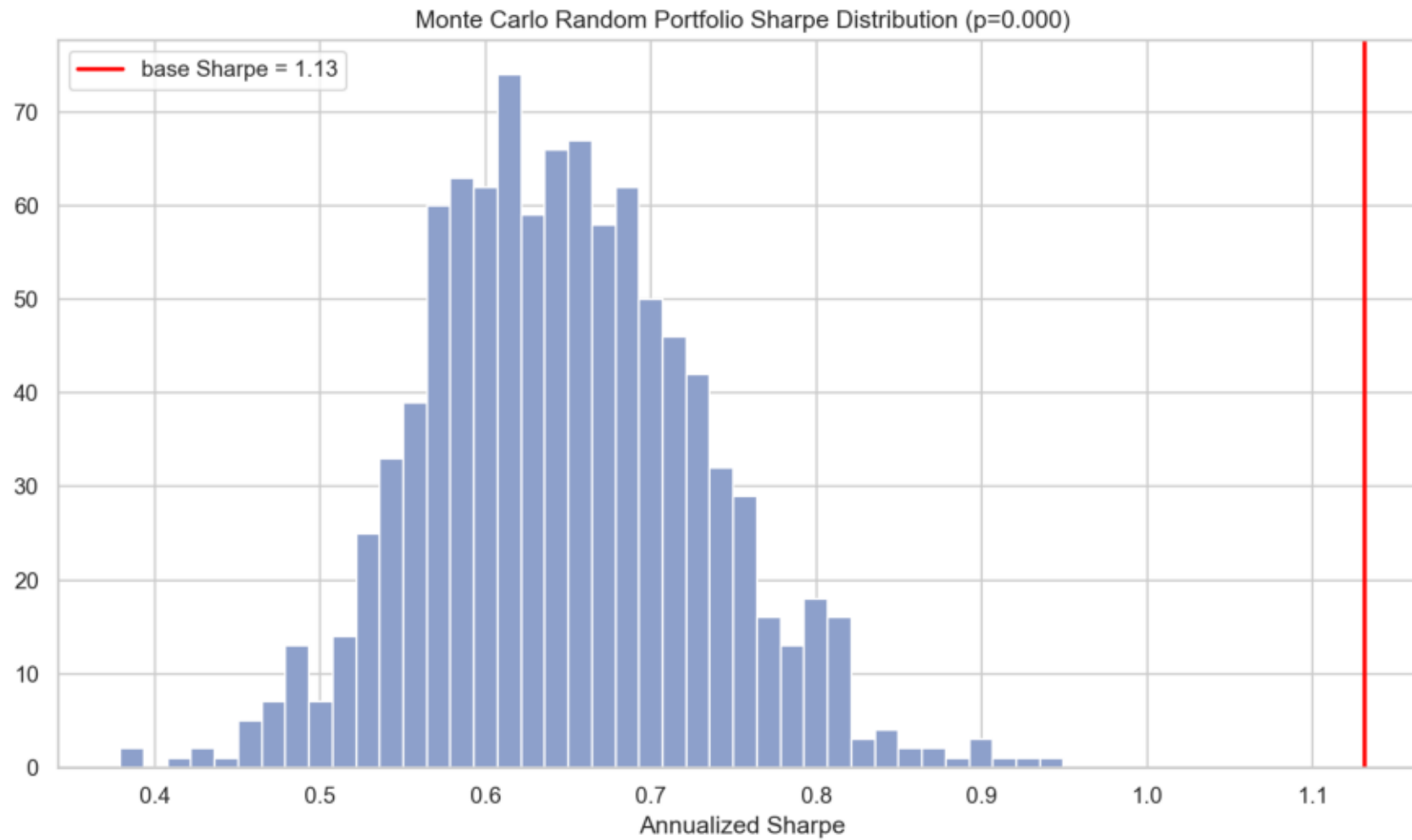
Drawdowns



Improvement Tests



Monte Carlo Robustness



Backtrader Results

- Base Backtrader final value: \$5,778,291; Sharpe: 1.20.
- Improved 3 Backtrader final value: \$2,680,533; Sharpe: 0.85.
- Settings: initial cash 1,000,000; FixedCashSizer 100,000; commission 0.
- Improved 2 and improved 3 use daily Backtrader `bt.Order.Stop` / `bt.Order.Limit` protective exits.
- Saved outputs include orders, trades, positions, equity curves, and metrics.

Main Results

- Base strategy final equity: \$5,319,947; Sharpe: 1.13.
- Improved 3 strategy: improved_3_dynamic_ic_weights_stop_take_top10; final equity: \$2,635,221; Sharpe: 0.68.
- Monte Carlo p-value for base Sharpe: 0.0000.
- Monte Carlo p-value for improved 3 Sharpe: 0.1710.
- Improved 3 annualized alpha vs ^GSPC: 5.43%.
- Interpret results cautiously because the universe has survivorship bias and costs are ignored.

Conclusion

- The factor process is economically interpretable and reproducible.
- The strongest evidence comes from comparing base, improvement, walk-forward, and random-portfolio tests.
- We would not present this as real-money ready without survivorship-bias-free data and realistic costs.
- Future work: historical S&P membership, transaction costs, slippage, and stronger out-of-sample validation.